

M-1977-208

1697 Cole Boulevard  
Suite 200  
Golden, CO 80401

Tel: (303) 239-5400  
Fax: (303) 239-5454

RECEIVED

MAR 28 2006 ✓

Division of Minerals and Geology

✓  
BROWN AND  
CALDWELL

March 27, 2006

Ms. Erica Crosby  
Division of Minerals and Geology  
Department of Natural Resources  
1313 Sherman Street, Room 215  
Denver, Colorado 80203

Re: Brown and Caldwell Presentations ✓  
CEMEX C-Pit Meeting, March 22, 2006 ✓

Dear Erica:

Please find enclosed on a CD-ROM copies of the two presentations that we discussed at our meeting last Monday in your offices. These presentations address the analytical issues surrounding C-Pit and nearby groundwater, as well as the treatability issues associated with TR-07.

If you have any questions please don't hesitate to contact me.

Sincerely,

BROWN AND CALDWELL



Al Notary, P.E.  
Managing Engineer

cc: Eric Hodek, CEMEX  
Paul Banks, Banks & Gesso

# **Explanation for Discrepancies in Monitoring Well Selenium Data**

March 22, 2006

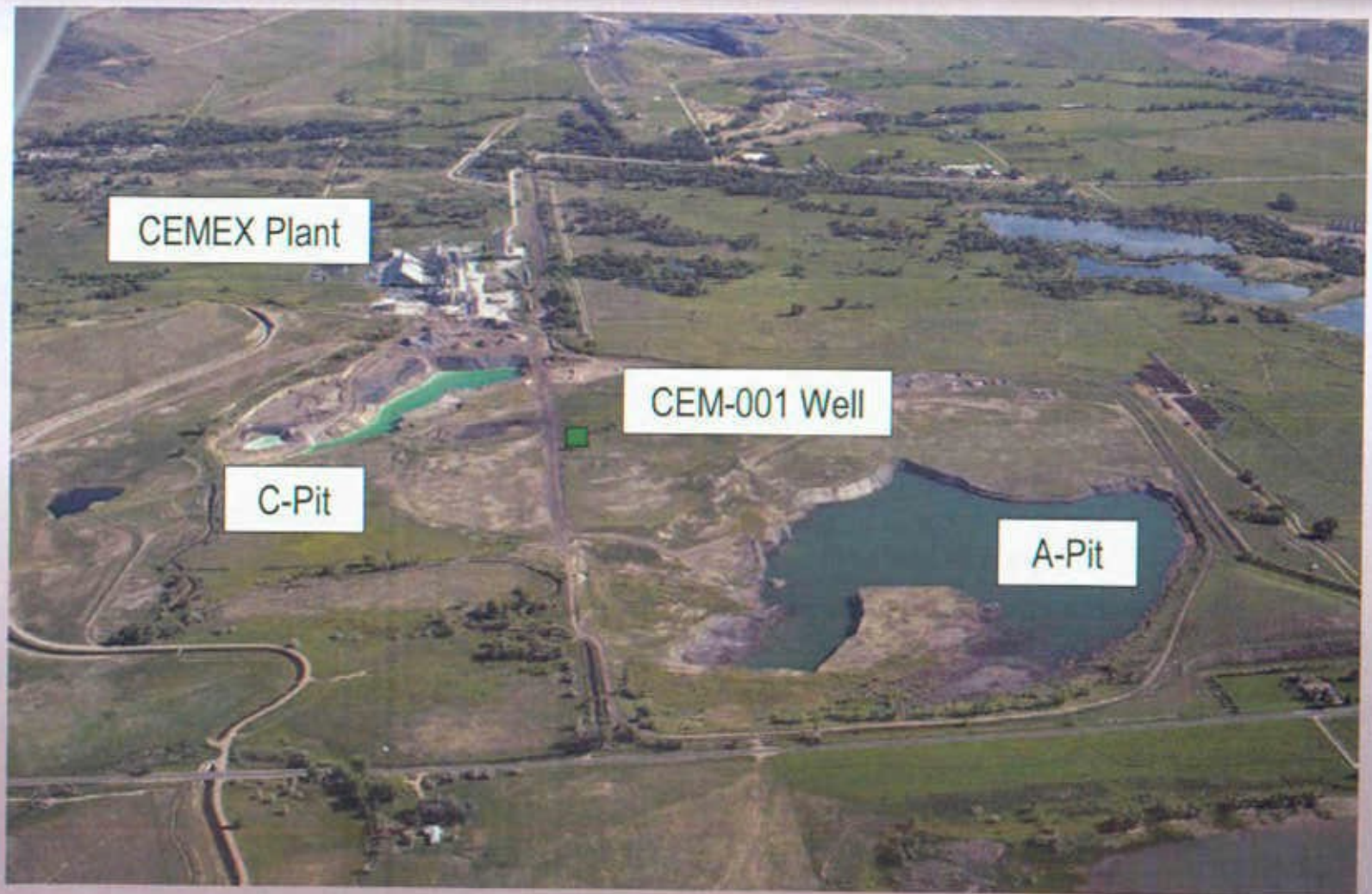
## **CEMEX Lyons Facility**



BROWN AND  
CALDWELL



# Location of Monitoring Well





# Selenium Reported in Monitoring Well

| Sample date | Selenium, mg/L |            |
|-------------|----------------|------------|
|             | ICP-MS         | ICP-DRC-MS |
| 7/22/05     | 0.0011         | NA         |
| 12/13/05    | 0.029          | NA         |
| 1/19/06     | 0.027/<0.001*  | NA         |
| 2/17/06     | 0.082          | 0.0034     |

\*Two laboratories

NA = Not analyzed

ICP-MS: Inductively-Coupled Plasma Mass Spectrometry

ICP-DRC-MS: Inductively-Coupled Plasma Dynamic Reaction Cell Mass Spectrometry

# Principles of MS

- MS = Mass Spectrometry
- A mass is selected, and isotopes at that mass number are counted
- Concentration at that mass number is divided by natural isotopic percentage



# Principles of MS, continued

Selenium:

- 9 %  $^{76}\text{Se}$
- 8%  $^{77}\text{Se}$
- 24 %  $^{78}\text{Se}$
- 50 %  $^{80}\text{Se}$
- 9 %  $^{82}\text{Se}$

# Selenium Determination

- Determine concentration of species at mass 82
- Assume it is all  $^{82}\text{Se}$
- Divide result by 9% to get total selenium concentration

Incorrect  
assumption!



# Bromine Hydride Interference

- $^{81}\text{Br}$  is 50% of all bromine
- In ICP-MS, some can form hydride  $^{81}\text{Br}^1\text{H}$ , which has mass 82
- If sample contains significant bromide, get positive interference for  $^{82}\text{Se}$
- ICP-DRC-MS eliminates hydrides



# Bromide in Monitoring Well

- Not a routine analyte
- 2/17/06 contained 20 mg/L bromide
- Less than 0.04% of bromide would have to form  $^{81}\text{Br}^1\text{H}$  to cause the 0.082 mg/L selenium reading

# 2<sup>nd</sup> and 3<sup>rd</sup> QTR 2006 Sampling

- Selenium by both methods
- Bromide



# Options for C-Pit Water Treatment

March 22, 2006

## CEMEX Lyons Facility



BROWN AND  
CALDWELL

# C-Pit Water Options

- Neutralize and use in cement process (process reuse option)
- Treat and apply to land



# C-Pit and A-Pit: July 2004



# Treatment Processes Tested

- Ferrous iron reduction
- Biological reduction



# Treatment Efficacy Depends on Selenium Speciation

Possible selenium species:

- Selenite  $\text{SeO}_3^-$  and selenate  $\text{SeO}_4^-$
- Elemental selenium  $\text{Se}^0$
- Organic selenium species  
e.g. dimethylselenide  $(\text{CH}_3)_2\text{Se}$   
and selenocysteine  $\text{C}_3\text{H}_7\text{NO}_2\text{Se}$

# Selenium Speciation

| Sample date | Concentration, $\mu\text{g/L}$ |                              |          |
|-------------|--------------------------------|------------------------------|----------|
|             | Selenite<br>$\text{SeO}_3^-$   | Selenate<br>$\text{SeO}_4^-$ | Total Se |
| 11/4/2005   | 31.8                           | 401                          | 453      |
| 2/16/2006   | 40.5                           | 368                          | 468      |

Analyzed by Applied Speciation and Consulting, LLC, Tukwila, Washington



# Selenium Speciation

| Sample date | Concentration                |          |
|-------------|------------------------------|----------|
|             | Sum of selenite and selenate | Total Se |
| 11/4/2005   | 433                          | 453      |
| 2/16/2006   | 409                          | 468      |

$$\Delta = 20 \mu\text{g/L}$$

$$\Delta = 59 \mu\text{g/L}$$

Analyzed by Applied Speciation and Consulting

# Selenium Speciation

- Some species other than selenate and selenite may be present
- Other species may not be removeable

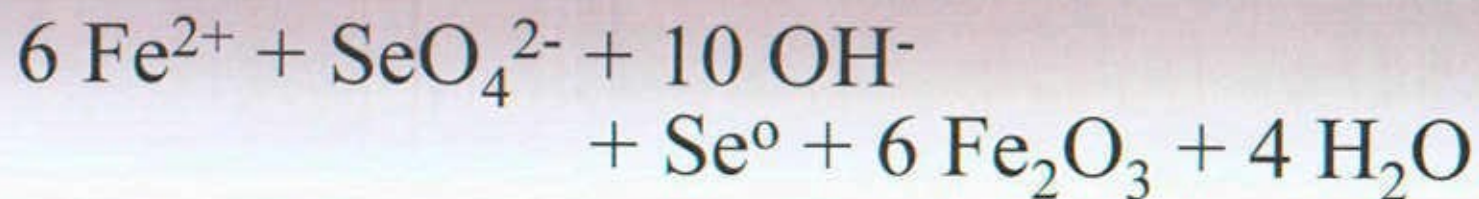
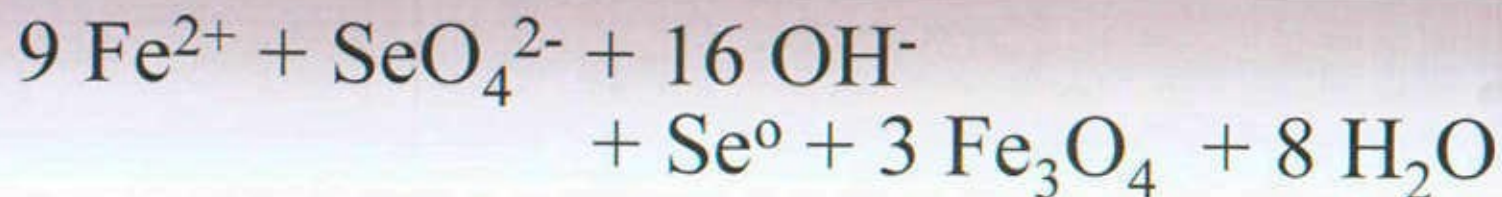


# Treating for Selenium

Two selenium removal processes tested:

- Ferrous iron ( $\text{Fe}^{2+}$ )  
reduction/precipitation
- Biological reduction

# Ferrous Reduction/Precipitation



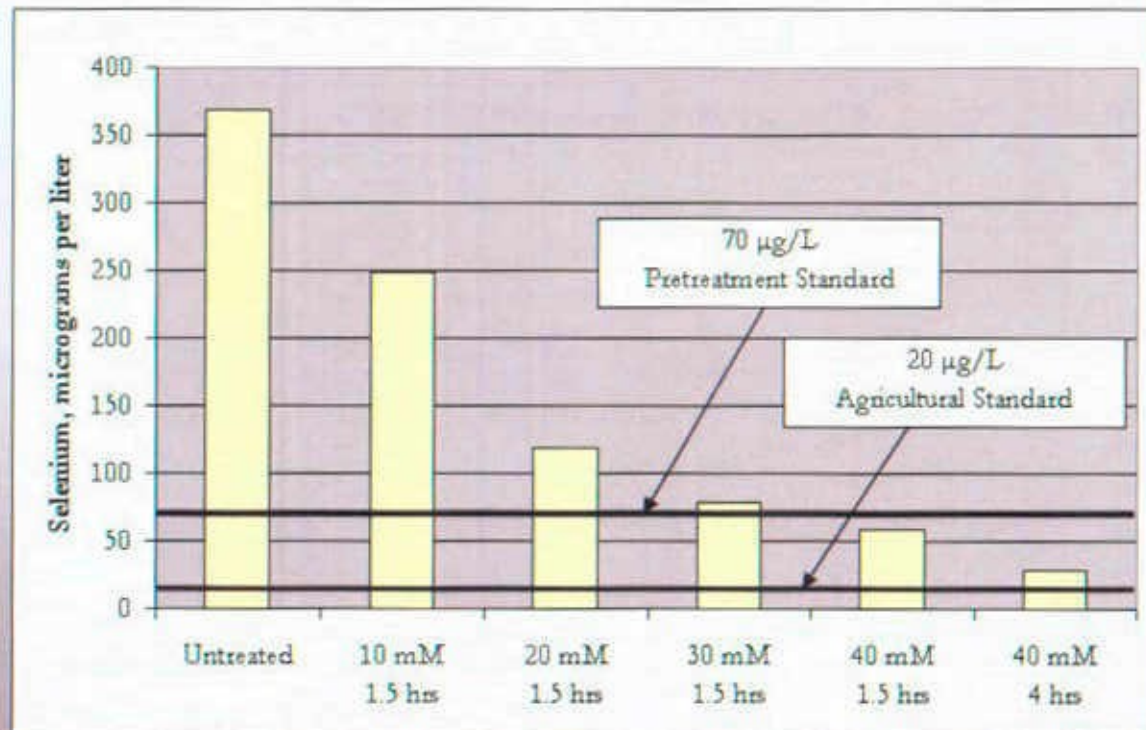


# Ferrous Reduction/Precipitation

- Adjust to pH 9 with acid
- Add ferrous chloride
- Mix 1.5 - 4 hours
- Settle to remove sludge

# Ferrous Reduction/Precipitation

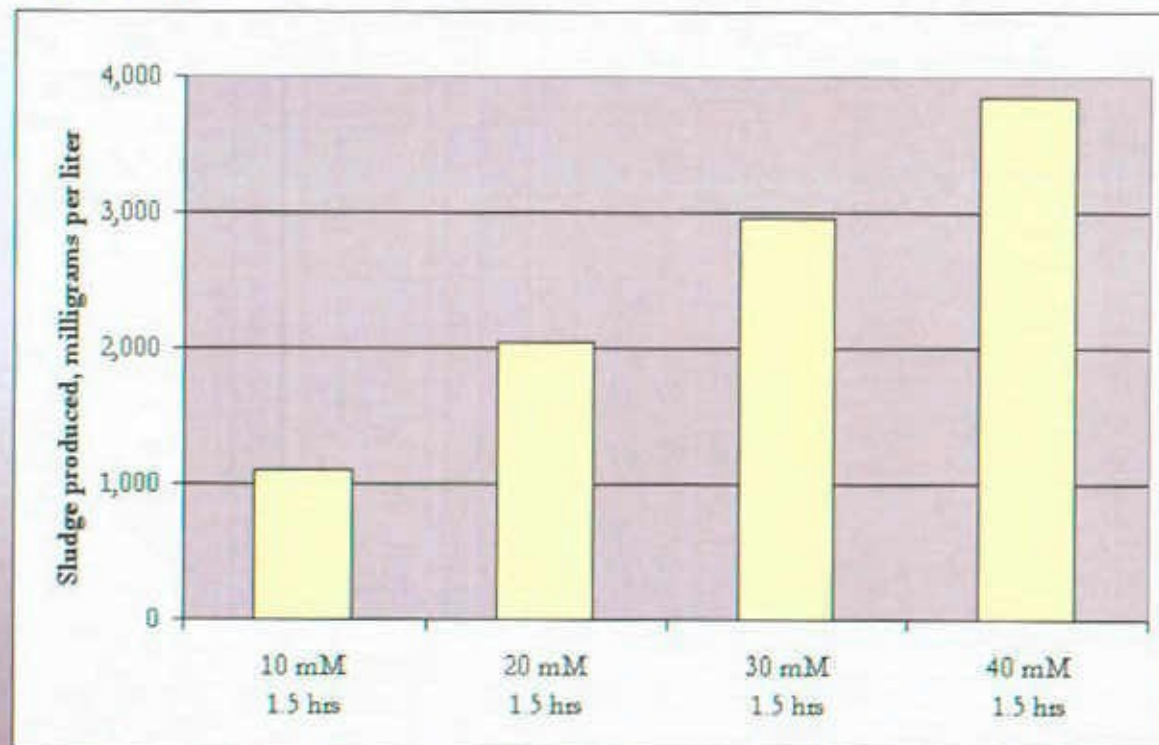
Selenium removal:





# Ferrous Reduction/Precipitation

Sludge production:



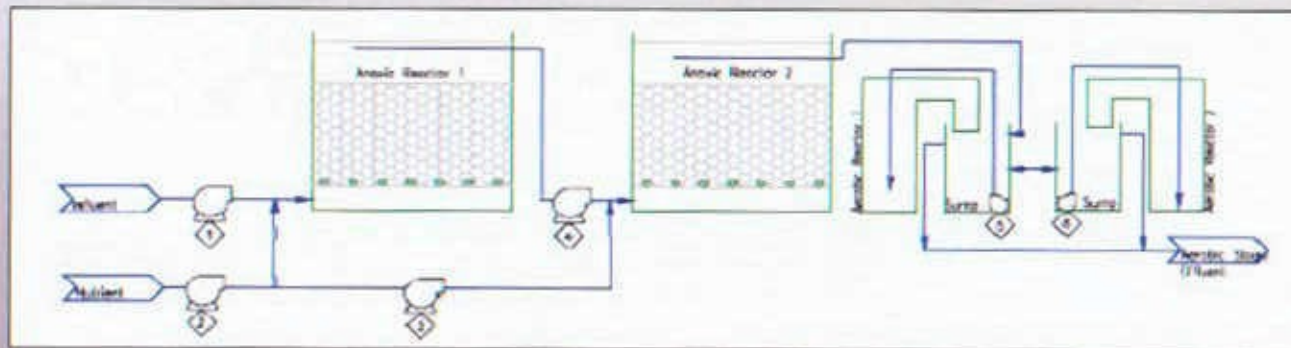
# Biological Reduction

- Certain microbes reduce selenate and selenite to elemental selenium
- Carbon source must be added
- Oxygen must be excluded (anoxic)
- Analogous to denitrification (nitrate reduction)

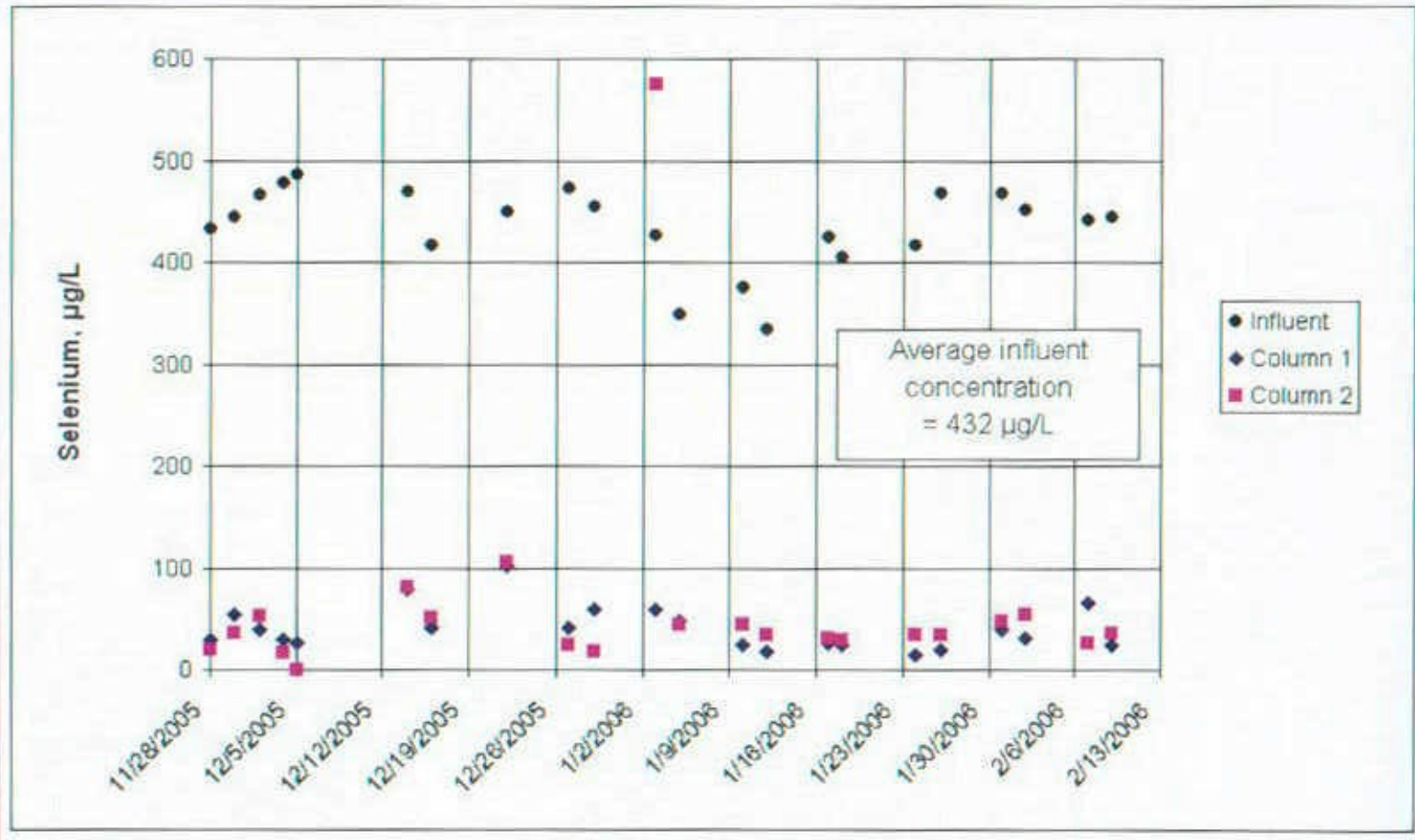


# Biological Reduction

- Zenon Environmental has packed bed process using activated carbon
- Similar waters have been treated to  $< 10 \mu\text{g/L}$  selenium

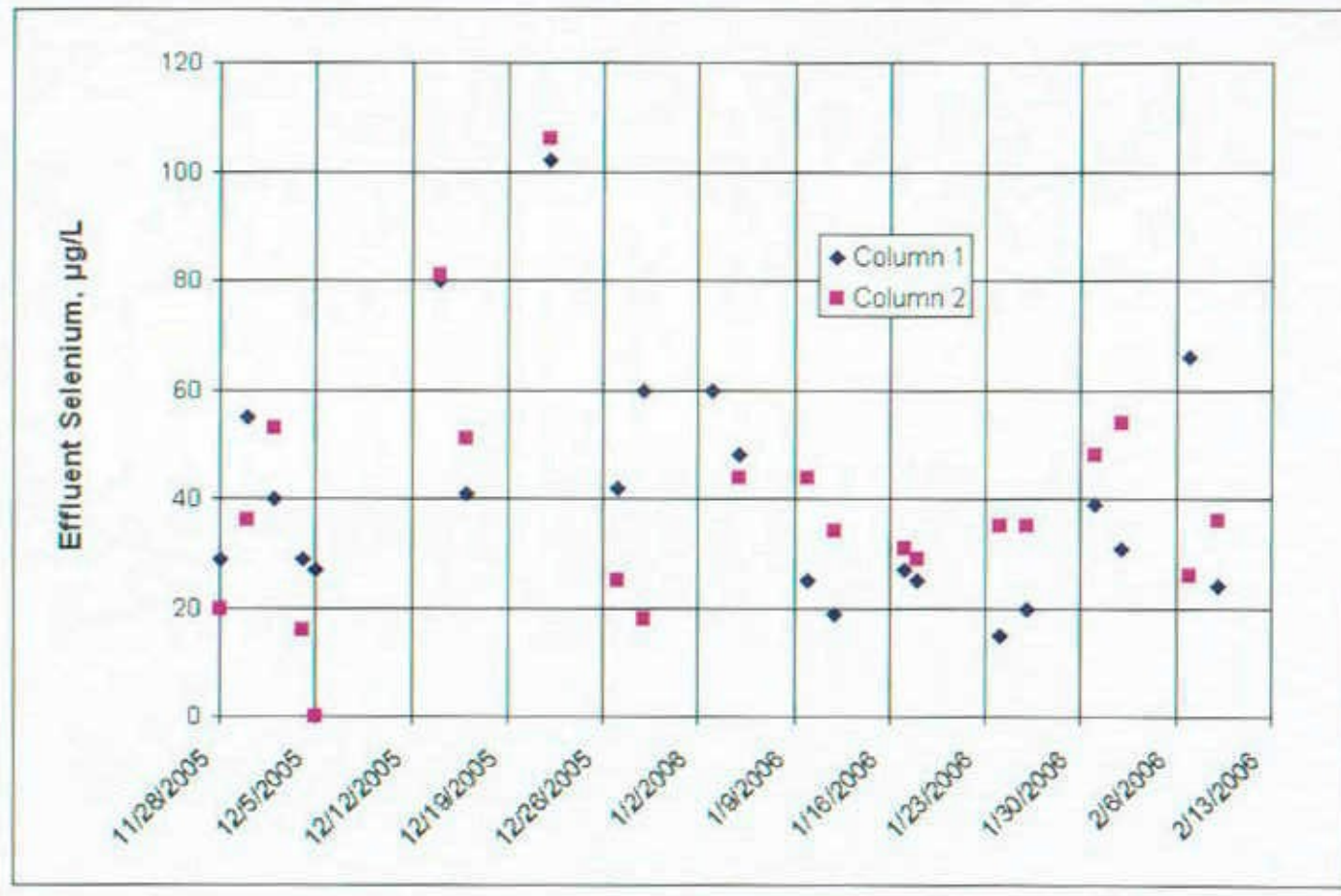


# Biological Reduction





# Biological Reduction



# Biological Reduction

## Conclusions:

- Biological reduction removed 90%
- Effluent remains at 30  $\mu\text{g/L}$  to 50  $\mu\text{g/L}$
- May be due to non-reducible species in C-Pit water



# Treatment Study Conclusions

- Ferrous iron reduction and biological reduction each remove 80% to 90% of selenium
- Neither process was shown to achieve 20 µg/L agricultural standard

# Action Plan

- Per TR-6, CEMEX will consume C-Pit water in the process commencing in April 2006.
- CEMEX will submit TR-7 for bonding purposes, based upon the biological reduction option. Treated water would be used for reclamation and dust control.